Serverless Daily Task Scheduler with Automated Email Reminders on AWS

1 Frontend (User Interface)

Web application for managing daily tasks

User (Web Browser)

HTTPS

AWS Amplify Hosting

HTML / CSS / JavaScript frontend

User Actions

- View all tasks
- Add new task with reminder time
- Mark task as complete

2 API Layer

Exposes REST API endpoints

Amazon API Gateway

REST API (HTTPS Endpoints)

API Endpoints

POST	/task	→ Create task
GET	/task	→ Fetch all tasks
PUT	/task	→ Update task status

3 Compute Layer (AWS Lambda Functions)

Business logic for task management and reminders

CreateTask Function

- Store task in DynamoDB
- Create one-time reminder schedule

GetTasks Function

- Fetch tasks from DynamoDB

UpdateTask Function

- Update task status in DynamoDB

Store task data (Read / Write)

Create one-time schedule

Update task data (Write)

4 Data & Scheduling Layer

Stores task data and manages reminder schedules

Amazon DynamoDB

DailyTaskScheduler table

- Stores task details
- PK: userID
- SK: taskID
- Task info (title, time, status, etc.)

Create Schedule (one-time)

Amazon EventBridge Scheduler

- Creates one-time schedule
- Triggers reminder at specified time

Invokes at scheduled time

5 Notification Layer

Sends email reminders to users

ReminderFunction

- Fetch task details from DynamoDB
- Format email message
- Publish message to SNS

sns:Publish

Amazon SNS

- Sends email notification
- Topic with email subscription

User Email

(Email Subscriber)

Receives task reminder email

6 Security & Monitoring

Secure access and operational visibility

AWS IAM

- Least-privilege permissions
- Lambda access to DynamoDB, EventBridge Scheduler, SNS
- API Gateway execution role

IAM roles and permissions

Amazon CloudWatch

- Lambda function logs
- API Gateway logs
- Monitoring and troubleshooting

Send logs

End-to-End Flow

From creating a task to receiving an email reminder

1. User opens the web app hosted on AWS Amplify.
2. Frontend calls API Gateway endpoints over HTTPS.
3. Lambda functions create, fetch, or update task data in DynamoDB.
4. CreateTaskFunction creates a one-time schedule in EventBridge Scheduler.
5. At the reminder time, ReminderFunction sends an email through SNS.